XISD6329 — TASK 1, Group 13

Work Integrated Learning 3B

Updated Project Plan, Site Map & Wireframes

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SAFEZONE SACOMMUNITY SECURITY & SMART INCIDENT REPORTING PLATFORM

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Update Project Plan

Project Background &Update

SafeZone SA is a community security and smart incident reporting platform. The project analysis and design into implementation, testing, version control and deployment. The solution retains incident reporting, moderation, assignment, tracking, reporting and audit concepts while adding a mobile application and a DevOps-oriented development lifecycle.

Client Needs and Problem Description

SafeZone SA addresses the need for a structured digital channel through which community members can submit safety incidents and monitor their progress. Analysis identified the need for accessible reporting, better visibility, structured moderation, and centralized incident information. The solution provides web and mobile access while maintaining privacy, security, role-based access and accountability.

Project Scope

In scope:

- Responsive website for public and authenticated users.
- Android mobile application for incident reporting and tracking.
- Registration and login.
- Community dashboard and incident submission.
- Incident status tracking.
- Moderator review, approval/rejection and assignment.
- Administrator user management and reporting concepts.
- PHP API used by the mobile application.
- MySQL database for users, incidents, locations, media, assignments and audit records.
- Azure DevOps Kanban lifecycle management and testing evidence.

Project Objectives

- Implement a functional Android mobile application using the SafeZone SA API.
- Connect both clients to the same database-backed service.
- Apply validation and role-based access to the prototype.
- Use Azure DevOps/Kanban to manage the application lifecycle.
- Create and execute appropriate unit/functional tests.
- Prepare installation and help information

Team Roles and Deliverable Allocation

Team Member	Role	Responsibilities	Deliverables
Nkuna Kuhlula	Project Manager / DevOps	Coordination, sprint planning, Azure DevOps/Kanban, progress tracking	Project plan, Azure evidence, repository coordination
Gqoboka Lukhanyo	Software Developer / Development Lead	Website, backend and API implementation	Website pages, PHP/API components
Oladele Oladayo	Security Champion / Business Analyst	Requirements, security, POPIA-aware design, documentation	Analysis, security controls, documentation
Muti Mulweli	Software Developer / QA Lead	Mobile development, testing and QA	Android screens, tests, QA evidence

DevOps Lifecycle

The development approach follows a continuous lifecycle in which planning, development, testing, release and feedback inform the next iteration. Azure DevOps is used for lifecycle and Kanban management.

Stage	SafeZone Activity	Tool / Evidence	Output
Plan	Backlog, prioritization and work allocation	Azure DevOps Kanban	Tasks / priorities
Code	Website, API, database and Android development	GitHub / Android Studio / VS Code	Source code / commits
Build	Compile and validate project components	Android Studio / local environment	Build artefacts
Test	Unit/functional testing and defect correction	Azure DevOps + test evidence	Results / fixes
Release	Prepare stable demonstration version	GitHub versioning	Release candidate
Deploy	Run website/API and install Android prototype	XAMPP / hosting / emulator	Running prototype

Operate	Observe behavior and resolve issues	Logs / issue tracking	Operational feedback
Feedback	Use feedback to update backlog	Azure DevOps	Improvement tasks

Development Technologies and Tools

Area	Technology / Tool	Purpose
Website	PHP, HTML, CSS, JavaScript	Web interface and server-side processing
API	PHP REST-style endpoints	Communication between Android and database-backed service
Mobile	Android Studio, Kotlin, Retrofit	Android client and API communication
Database	MySQL	Persistent data storage
Source Control	GitHub	Version control
Lifecycle	Azure DevOps Kanban	Backlog, tasks and lifecycle management

Project Deliverables

- Updated project plan.
- Website and mobile application site maps.
- Website and mobile wireframes.
- Working website prototype.
- Working Android mobile prototype.
- Connected database and API prototype.
- GitHub repository with README and organized assets.
- Azure DevOps Kanban evidence.
- Unit/functional testing evidence.
- Installation/help instructions.

Project Timeline

Phase	Activities	Evidence	Responsible
1. Planning	Review Semester 1; update scope; backlog; allocation	Updated plan / Kanban	All / PM
2. Design	Update sitemap and web/mobile wireframes	Sitemap / wireframes	BA + Developers
3. Website & DB	Implement website, database and server functions	Website / database	Development
4. API & Mobile	Implement PHP API and Android Retrofit integration	API / Android	Development
5. Testing	Execute tests and fix defects	Test evidence	QA + team
6. DevOps & Release	Commits, Kanban updates, demo build	GitHub / Azure evidence	PM + team
7. Presentation	Prepare slides and live demonstration	Presentation / demo	All

Risks, Controls and Recommendations

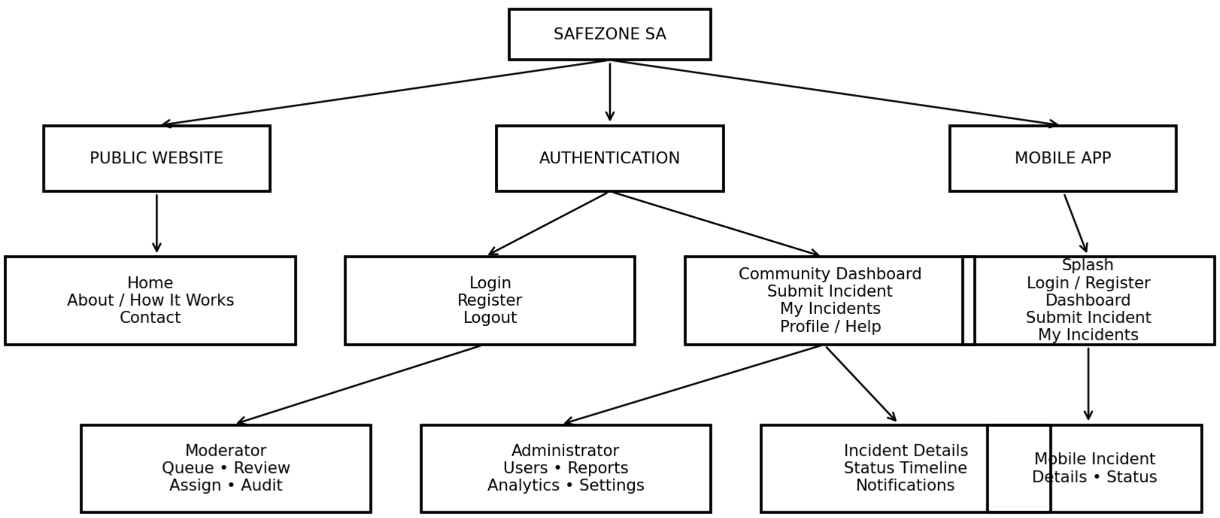
Risk	Impact	Mitigation	Owner
API/mobile connectivity failure	High	Verify emulator networking, API base URL and endpoints.	Development / QA
Database inconsistency	High	Use defined relationships, validation and repeatable SQL setup.	Development
Security/privacy weakness	High	Minimize PII, validate input, use RBAC and audit information.	Security Champion
Scope creep	Medium	Prioritize MVP backlog and manage changes through Azure DevOps.	PM
Version conflicts	Medium	Use regular GIT commits and coordinated changes.	All
Weak evidence	Medium	Capture screenshots of key workflows and prepare a timed demo.	All

Site Map

The SafeZone SA navigation is organized around public access, authentication and role-based functions. The mobile application mirrors the core community workflow while keeping navigation suitable for smaller screens.

SAFEZONE SA- WEBSITE & MOBILE SITE MAP

Figure 1: Website and Mobile Site Map

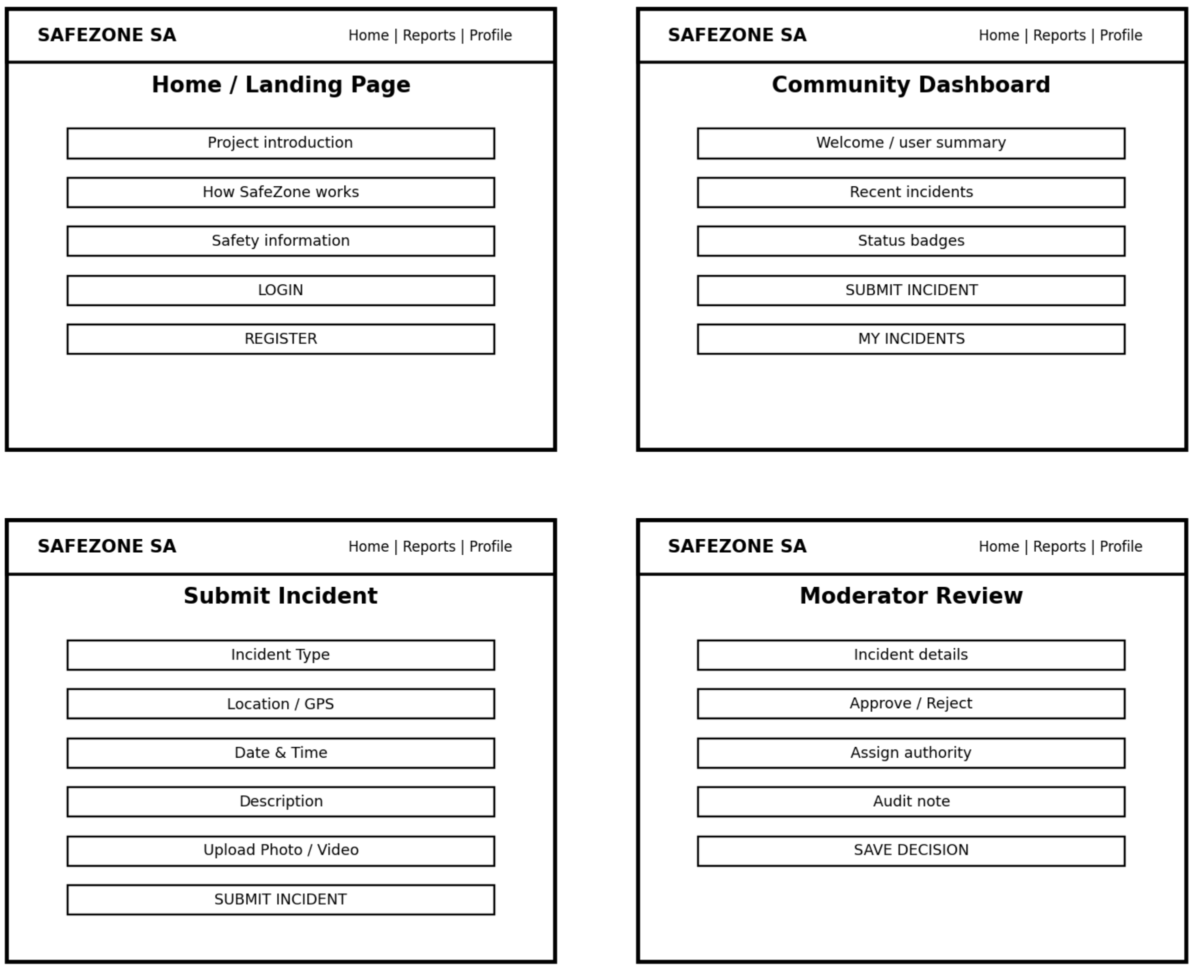


Wireframes

The wireframes translate the requirements and site map into a visual prototype before final implementation. They prioritize clear navigation, incident reporting, status visibility and role-specific functions.

Website Wireframes

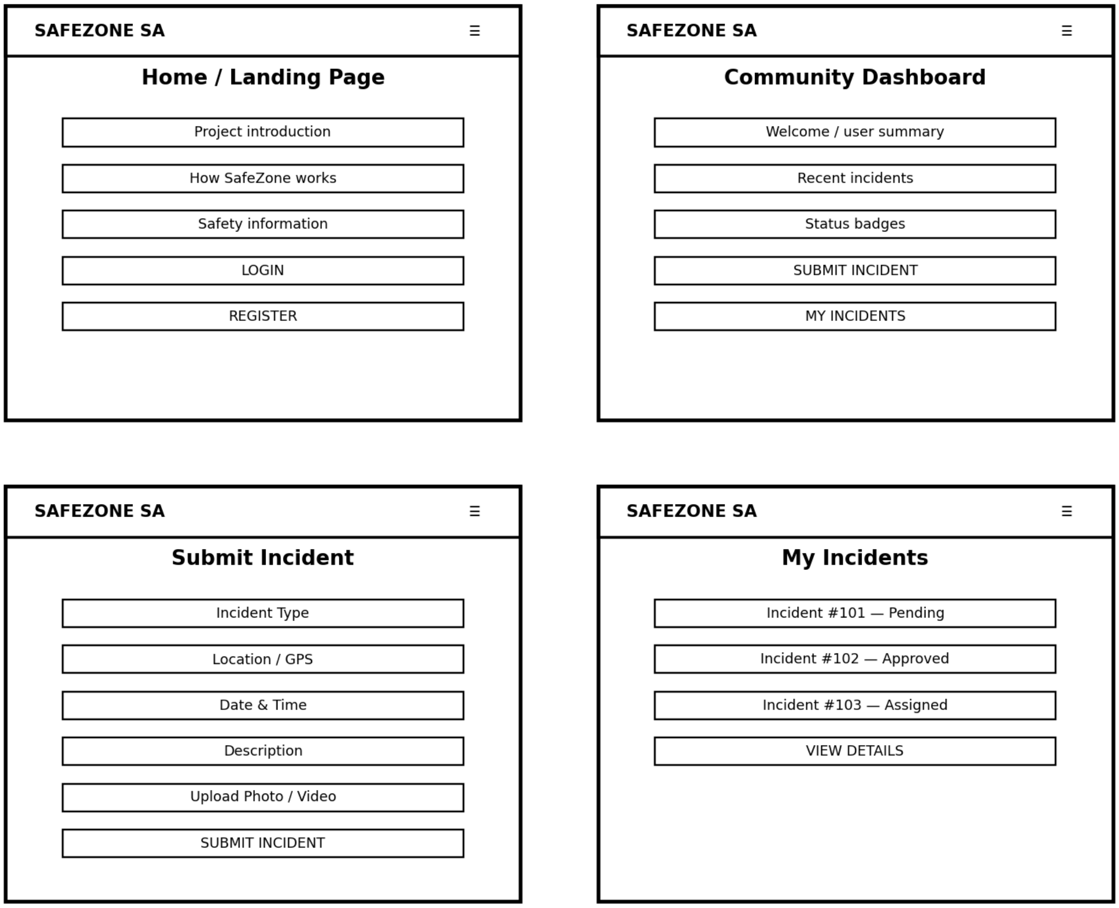
Figure 2: Website Wireframes



The website wireframes cover the public landing page, community dashboard, and incident submission and moderator review workflow.

Mobile Application Wireframes

Figure 3: Mobile Application Wireframes



The mobile wireframes cover authentication, the community dashboard, incident submission and incident tracking. The Android application is intended to use the same backend/API and database as the website

References

https://github.com/ST10157844/SafeZone-SA.git